# Introduction to Programming

## Lecture 18

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# File

# Types of Files

- Text Files
- Executable Programs

- Memory is volatile
- Any data that you key in by keyboard while a program is running is also volatile

# File Handling

- Text files handling
- Binary files handling

# Steps to handle file

- Open the File
- Read / Write the File
- Close the File

# Streams

# Header File for File Handling

# fstream.h

# Header File for File Handling

```
#include <fstream.h>
```

# Input File Stream

ifstream

# Output file stream

ofstream

# Example 1

```
#include <fstream.h>

ifstream myFile ;

myFile.open ( “payRoll.txt” ) ;
```

# Fully Qualified Path Name

C:\myProg\payRoll.txt

# Access file data

```
myfile >> var1;
```

We can also write:

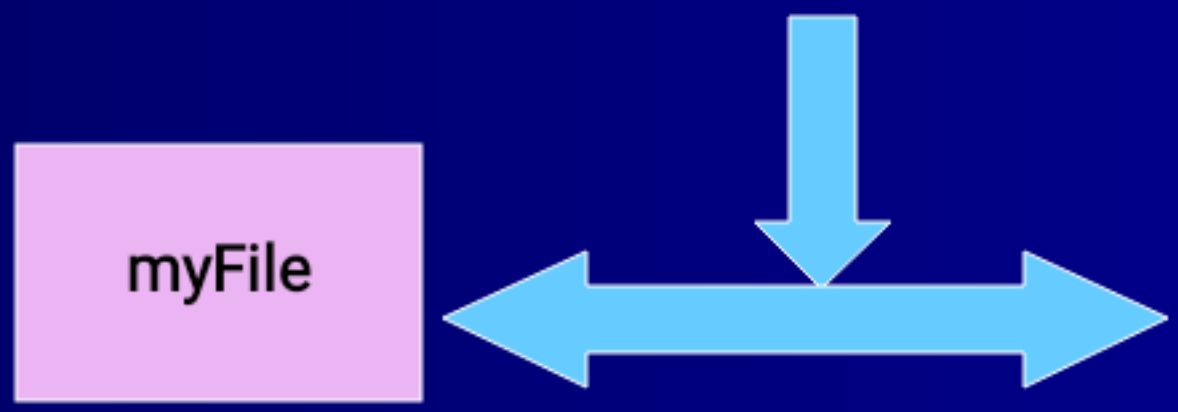
```
myfile >> var1 >> var2 ;
```

# Close the File

```
myFile.close ( ) ;
```

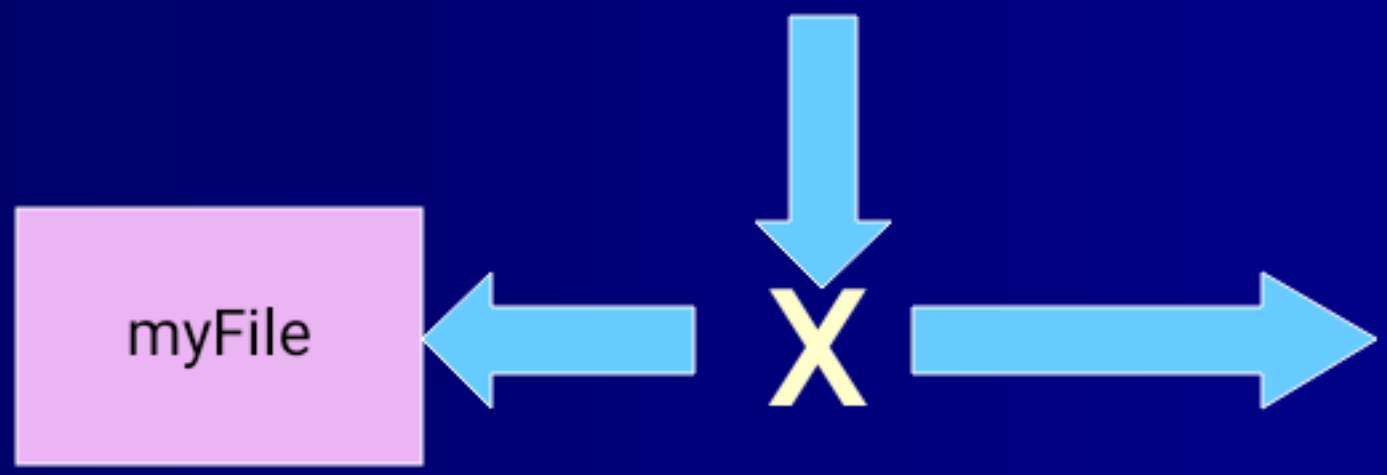
# Process : Open

myfile.open ( “payRoll.txt” ) ;



# Process: Close

myfile.close ( “payRoll.txt” ) ;



# Example 1

```
ifstream myFile ;
myFile.open ( “myFile.txt” ) ;

if ( !myFile )  // Error check
{
 cout << “Your file could not be  opened”;
}
 ------
myFile.close ( ) ;
```

# Output File Modes

- Create a new file
- Overwrite an existing file
- Append some text
- Randomly accessing a file

# Syntax

fStream fileVar ( “fileName” , mode ) ; // Generic syntax

Opening Mode

ifstream myfile ( “myfile.txt” , ios :: in ) ;

Opening Mode

ofstream myfile ( “myfile.txt” , ios :: out ) ;

# List of File Handling Modes

ios :: in open for reading (default for ifstream)

ios :: out open for writing (default for ofstream)

ios :: app start writing at end of file (APPend)

ios :: ate start reading or writing at EOF of file (ATEnd)

ios :: trunc truncate file to zero length if it exists (TRUNCate)

ios :: nocreate error when opening if file does not already exist

ios :: noreplace error when opening for output if file already exists

ios :: binary open file in binary (not text) mode



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- Append
  ofstream myfile (“myfile.txt” , ios :: app ) ;
- Random Access
  ofstream myfile ( “myfile.txt” , ios :: ate ) ;
- Truncate
  ofstream myfile ( “myfile.txt” , ios::trunc ) ;

# myfile.eof ( )

```
while ( !myfile.eof ( ) )
{
 myfile >> varName ;
}
```

# get ( )

```
char ch ;
myFile.get ( ch ) ;
```

# Example 2

```
while ( !myFile.eof ( ) )
{
  myFile.get ( ch ) ;
  cout << ch ;
}
```

# put ( )

```
outputFile.put ( ch ) ;
```

```
ifstream myInputFile ( “myfile.txt” , ios :: in ) ;

ofstream myOnputFile ( “myfile.txt” , ios :: out ) ;
```

```
int i ;
i = 0 ;
```

```
int i = 0 ;
```

# Open file

```
ifstream myInputFile ( “myfile.txt” ) ;
```

# Open file

```
ofstream myOnputFile ( “myfile.txt” ) ;
```

# strtok ( )

# getline ( ) function

# Syntax of getline function

Character array

Numbers of characters to be read

myfile.getline (char *s, int n, char delim);

delimiter

# Syntax of getline function

Input

Hello World

myfile.getline ( arrayString , 20 , ’W’ ) ;

Output

Hello

# File

Amir 1000

Amara 1002

# strtok ( string , delimiter )

# Example

```
char * namePtr , *salaryPtr , arr [ 30 ] ;
double fSalary = 0.0 ;
inFile.getline ( arr , 30 , '\n' ) ;
namePtr = strtok ( arr , " " ) ;
salaryPtr = strtok ( NULL , " " ) ;
fSalary = atof ( salaryPtr ) ;
  ⋮
```